

Establishing a Multi-Dimensional Risk Framework over the Full Lifecycle of Stablecoins

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Abstract—With the rapid expansion of the crypto-asset market, stablecoins have emerged as the foundational infrastructure of the decentralized finance (DeFi) ecosystem. Concurrently, their endogenous financial risks and cross-market contagion effects have become pivotal issues within the domain of global financial regulation. This paper begins by tracing the evolutionary path, core theoretical foundations, and diverse application scenarios of stablecoins. It then defines their dual risk attributes, deconstructs 55 discrete atomic risks spanning the entire operational flow. On this basis, the paper constructs a two-dimensional classification framework for stablecoin risks, systematically mapping atomic-level risks across both the chronological dimension of the full lifecycle and the spatial dimension of amplification tiers. This mapping uncovers the core patterns of dynamic risk transfer and multi-tiered amplification.

Index Terms—Stablecoins, Atomic risks, Full lifecycle, Risk amplification tiers

I. THE BACKGROUND AND CORE THEORETICAL FOUNDATIONS OF STABLECOINS

A. The Background of Stablecoins

Since its inception in 2009, the crypto-asset market has demonstrated substantial potential within the domains of decentralized and programmable finance. However, the endogenous limitations of native crypto-assets (such as Bitcoin and Ethereum) directly catalyzed the emergence of stablecoins. Native cryptocurrencies have historically exhibited pronounced price volatility. Evaluated through the lens of modern monetary economics, these severe price fluctuations impose profound constraints on their capacity to fulfill fundamental monetary functions, particularly in serving as a reliable unit of account and a medium of exchange [1].

B. The Evolutionary Trajectory of Stablecoins

The evolutionary trajectory of stablecoins traces a clear path from monolithic mechanisms toward diversification, and from crypto-native tools toward macro-financial infrastructure. The introduction of Tether (USDT) in 2014 marked the beginning of the stablecoin era. To mitigate the volatility that limited early crypto-assets as exchange media, USDT introduced a dollar-anchored architecture, backed by a purported one-to-one fiat reserve [2]. As the market expanded and institutional demands grew more complex, 2018 marked a critical turning point in the divergence of stablecoin mechanisms. The market witnessed not only the birth of USD Coin (USDC)—which leveraged regular audits and high reserve transparency as its

core competitive advantages—but also the rise of decentralized models, epitomized by MakerDAO’s DAI. Instead of relying on centralized financial institutions, DAI stabilizes its value through an on-chain system of crypto-asset over-collateralization.

Post-2020, the explosive expansion of decentralized finance (DeFi) radically reshaped the market positioning of stablecoins. Transitioning rapidly from mere unit-of-account and settlement instruments, stablecoins evolved into the foundational credit infrastructure of the DeFi ecosystem. They began serving as core collateral and primary liquidity media across lending protocols, decentralized exchanges (DEXs), and derivatives markets, thereby strengthening the interconnectedness between digital assets and the traditional financial system [2]. However, this unprecedented expansion coexisted with a compounding accumulation of systemic risks. The catastrophic collapse of the algorithmic stablecoin TerraUSD (UST) in 2022 marked a critical watershed moment for the industry. The UST crisis illuminated the profound fragility of stabilization mechanisms and effectively concluded the phase of disorderly expansion within the crypto market, catalyzing a substantive overhaul of the global regulatory architecture.

C. The Theoretical Foundations of Stablecoins

1) *Narrow Banking Architecture and Crypto Shadow Banking*: From a balance-sheet perspective, recent research from the Federal Reserve Bank of New York characterizes fiat-collateralized stablecoins as blockchain-native, narrow banking-like vehicles [3]. They pool public capital and allocate it almost exclusively toward ultra-liquid safe assets, such as cash and short-term Treasury bills. At the financial system, stablecoins effectively operate as the liquidity hubs of “crypto shadow banking.” By migrating their balance sheets on-chain, issuers evade conventional “regulatory taxes,” including capital adequacy constraints and deposit insurance premiums. They subsequently leverage DeFi protocols to facilitate highly leveraged lending operations at scale. Although this structural architecture enhances operational financial efficiency, it concurrently breeds heightened liquidity vulnerabilities and systemic run risks.

2) *The “No-Questions-Asked” (NQA) Principle and Co-ordination Games*: For a medium of exchange to transition from a speculative asset to true money, it must satisfy the NQA principle [4]. As a form of private money in the digital

age, stablecoins lack the institutional backstop of government deposit insurance (e.g., FDIC) and the liquidity support of a central bank acting as the Lender of Last Resort (LOLR); consequently, they can never fully achieve a true NQA status. This institutional deficit highlights the source of their endogenous vulnerability. Tracing back to the seminal Diamond–Dybvig bank run model [5], the par redemption of stablecoins depends heavily on a game of “beliefs” among investors. The behavior of token holders is inherently a classic coordination game [6]. When macroeconomic conditions or public signals deteriorate, the system can instantly shift from a par equilibrium to a run equilibrium.

D. The Technical Feasibility Foundations of Stablecoins

Stablecoins derive their technical feasibility from distributed ledger technology (DLT), which serves as the foundational infrastructure. Leveraging this network, stablecoin systems have evolved a highly automated, collaborative architecture. Aligned with the classification framework of [7], the core functional modules deployed upon the ledger can be conceptualized as a three-dimensional technical stack comprising the collateral structure, price oracles, and the smart contract engine. Within this framework, a multi-asset collateral structure establishes the system’s foundational credit. Oracles bridge the protocol with real-world markets by continuously feeding external price signals on-chain. Concurrently, the smart contract serves as the central execution engine; guided by oracle price feeds, it frictionlessly triggers minting, burning (mint/burn), or liquidation logic, thereby achieving autonomous price self-correction without centralized intervention.

The efficacy of the automated closed-loop mechanism generated by these three modules is heavily contingent upon the technical performance of the underlying DLT. The distributed ledger not only enforces the legal and physical confirmation of collateral rights but also serves as the exclusive network pipeline for smart contract operations and micro-arbitrage execution. Consequently, the throughput and congestion levels of the underlying blockchain directly dictate the frictional costs inherent in the stablecoin’s arbitrage mechanism. Empirical evidence from [8] substantiates this technological dependency: the migration of Tether (USDT) from the inefficient Omni network to Ethereum significantly reduced the temporal costs and technical friction of arbitrage trading, thereby catalyzing a substantial convergence in secondary market price deviations. This demonstrates that a distributed ledger characterized by high throughput and low latency is an indispensable technical prerequisite for ensuring the low-friction execution of smart contracts and sustaining the stablecoin’s price anchor.

II. STABLECOIN APPLICATION SCENARIOS

Stablecoins were initially conceptualized as a bridge linking traditional finance with decentralized networks, aiming to synthesize the high-efficiency transmission of digital currencies with minimized price volatility [7]. Today, the practical application scenarios of stablecoins have expanded far beyond

this foundational vision of a mere cross-border payment instrument; they are now deeply integrated into both the micro-liquidity dynamics of the crypto market and the macro-level interactions with the traditional financial system.

A. The Endogenous Infrastructure of the Crypto Market: Safe Havens, Liquidity, and Leverage Engines

Stablecoins constitute the foundational infrastructure of the crypto market, driving decentralized finance through three core functions. Initially serving as passive safe havens that offer minimal friction, they enable immediate risk hedging during market drawdowns by bypassing the transaction costs of converting to traditional fiat currencies [9], [10]. Beyond mitigating risk, stablecoins have evolved into the primary liquidity bedrock of the ecosystem, commanding approximately 45% of the total liquidity across decentralized exchanges. In this capacity, they resolve the market deficit of a native settlement asset with low volatility and facilitate atomic settlement driven by smart contracts [11], [12]. Ultimately, within decentralized networks, stablecoins transcend their identity as a payment medium to operate as an engine for leverage. Through recursive loops of collateralization and borrowing, conservative lenders and speculative market participants generate a self-contained loop of supply and demand that exponentially expands capital efficiency while compounding systemic risk exposure across the entire financial framework [13], [14].

B. External Penetration into Traditional Finance: Macro-Level Circulation, Retail Dilemmas, and Sovereign Competition

While adjusted on-chain volume reached \$4.0 trillion in May 2026 [15], these nominal metrics mask a profound utility paradox. Of the \$35 trillion annualized volume, non-payment financial activities dominate; true end-user payments represent merely \$390 billion, roughly 1% of network activity [16]. Faced with this limited penetration, blindly expanding access to low-stickiness retail users exposes an adoption-fragility paradox. Rather than structurally stabilizing the ecosystem, this expansion exacerbates the risk of stampede runs during economic shocks [17].

Despite these vulnerabilities, stablecoins serve as vital alternative stores of value in hyperinflationary economies. For instance, they account for approximately 43% of transaction volume in Sub-Saharan Africa, where users leverage dollar-pegged assets to hedge against severe currency depreciation [18]. However, their macro-level expansion faces intense sovereign competition. Spearheaded by the Bank for International Settlements, central banks advancing unified ledgers and tokenized deposits leverage the unique advantage of absolute settlement finality to structurally crowd out private stablecoins in the broader financial market.

C. The Frontier Potential Unique to Web3.0 and AI Agents

Stablecoins are expanding beyond human transactions to serve as the primary media for settlement in the emerging machine economy, driven by the convergence of artificial

intelligence and Web3 technologies. Traditional financial networks fail to support high-frequency interactions among AI agents due to rigid compliance frameworks and high transaction costs. By utilizing blockchain architecture, stablecoins grant these agents autonomous economic status by seamlessly unifying identity verification with payment execution [19]. Furthermore, enabled by specialized protocols and scaling networks, these agents can execute microscopic transfers near zero cost, making nano-payments for data and computing power economically viable for the first time [20], [21]. Finally, embedding stablecoins within smart contracts enables instantaneous clearing upon the delivery of digital assets; this process of atomic settlement eliminates counterparty default risks and game-theoretic frictions, anchoring trust across decentralized networks [11].

III. RISK CLASSIFICATION AND CHARACTERIZATION OF STABLECOINS

A. The Dual Risk Attributes

Stablecoins and traditional financial assets exhibit a high degree of similarity in their core financial attributes and endogenous risk mechanisms. On the asset side, they align closely with constant net asset value (CNAV) money market funds (MMFs), relying on highly liquid safe assets such as sovereign Treasury bills to maintain par value stability, thereby inheriting the corresponding credit and market risks [2]. The surge in capital outflows observed when a stablecoin breaks its par threshold aligns tightly with the panic contagion mechanisms triggered when traditional MMFs “break the buck” [22]. On the liability side, they replicate the core transactional attribute of commercial bank demand deposits, offering on-demand redemption at par. Issuers generate interest margins through liquidity transformation, substantively performing the functions of an unregulated bank and endogenously bearing run vulnerabilities homologous to historical private banknotes [4].

Notwithstanding these operational parallels, stablecoins diverge fundamentally from traditional financial assets across several dimensions. Regarding risk-return allocation, while traditional MMFs compensate investors for credit risk via yield distributions, dominant centralized stablecoin issuers appropriate the entirety of the returns generated by the underlying reserves. Token holders absorb the structural risks without receiving commensurate compensation, culminating in a stark risk-return asymmetry [23]. In terms of risk propagation, whereas legacy bank runs are typically driven by information asymmetry, the radical transparency of the blockchain inadvertently catalyzes “front-running” behavior during stablecoin de-pegging events. This compresses the run cycle from days into minutes, manifesting a unique “transparency paradox” [6].

At the infrastructural level, traditional financial systems rely on decentralized and stringently regulated architectures. Conversely, stablecoins depend heavily on centralized, third-party technological intermediaries, engendering monopolistic “single-point-of-failure” risks rarely observed in legacy frameworks [24]. Within payment and settlement modalities,

traditional payment networks can internally absorb dispute-resolution costs. In contrast, the rigid irreversibility of stablecoin transactions transfers the entire burden of recourse onto end-users; furthermore, their on-chain settlement lacks statutory recognition of settlement finality, introducing profound legal defects [25]. In addition, while traditional demand deposits are backed by government deposit insurance and central bank Lender of Last Resort (LOLR) mechanisms, stablecoins operate entirely outside these safety nets. In the event of issuer insolvency, token holders are relegated to unsecured creditors lacking preferential claims over the reserve assets, leaving them exposed to the catastrophic risk of total principal loss.

B. An Atomic Risk Framework Based on the Dual Attributes of Stablecoins

The preceding comparative analysis demonstrates that stablecoins, serving as a bridge between traditional macroeconomy and DeFi, are inevitably entangled in the operational logic of traditional macro-finance, particularly regarding their exposure to runs and maturity mismatches. Concurrently, in terms of their physical substrate, they are entirely exposed to the nuances of distributed ledger technology (DLT) and the rigidity of smart contract code. Consequently, their risk profiles exhibit highly complex, cross-border coupling characteristics. Guided by the dual attributes of stablecoin architecture, these risks can be systematically categorized into “common risks” inherent to general financial assets and “unique risks” specific to special financial assets.

Building upon existing literature and the dual nature of traditional and crypto-native finance, this section examines the relevant entities involved across the entire operational flow from issuance to circulation. Through this lens, the complete lifecycle risks of stablecoins are deconstructed into 55 mutually independent “atomic risks,” which are indivisible risks at the root source of systemic vulnerability. Due to strict space constraints, the comprehensive taxonomy and detailed definitions of these 55 atomic-level risks, along with the full list of references, are provided in our anonymized supplementary material, available at: https://osf.io/mx4yf/overview?view_only=9327f99d3074408482ae8beb99651ee7.

The risk contagion of stablecoins represents a systemic evolutionary process triggered by either external macroeconomic shocks or underlying technological infrastructure failures. This contagion is amplified through the endogenous mechanisms of the asset side and the credit hub, erupts prominently within secondary markets, and ultimately propagates cross-market into the broader crypto ecosystem and traditional financial systems. Its core transmission pathway can be delineated into four progressive evolutionary stages (see Fig. 1):

C. The Risk Classification Framework of Stablecoins

Premised on the systematization of the 55 discrete atomic risks and the comprehensive analysis of the full-chain evolutionary pathway—spanning from initial risk triggers to global,

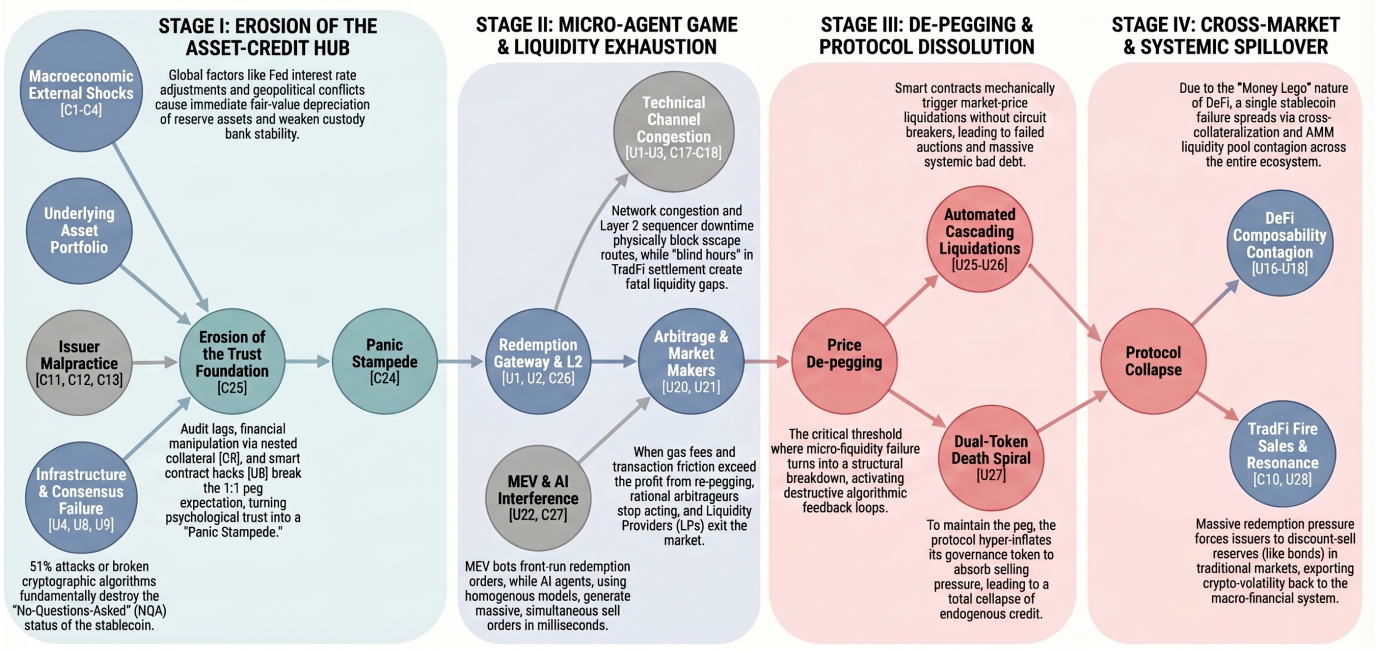


Fig. 1. The Risk Transmission Pathway of Stablecoins. (Note: This figure was illustrated with the assistance of Large Language Models (LLMs).)

cross-market contagion—this section constructs a temporal-spatial dual-dimensional classification framework for stablecoin risks. This framework executes a systematic and exhaustive mapping of the 55 atomic risks across both the temporal dimension (the full lifecycle) and the spatial dimension (risk amplification tiers). The horizontal chronological evolution along the temporal dimension, coupled with the vertical multi-tiered amplification along the spatial dimension, synergistically establishes a complete logical closed-loop that reveals stablecoin risk dynamics.

1) *Taxonomy of Stablecoin Risks: A Full Lifecycle Perspective*: Along the temporal dimension, oriented around the capital flow phases within the stablecoin lifecycle, this study utilizes the occurrence of credit derivation and leverage amplification as the foundational criterion to delineate between T1 and T2. Consequently, the complete operational cycle of a stablecoin is partitioned into four progressive stages: issuance and provisioning, circulation and payment, derivation and scenario expansion, and exit and liquidation. This framework precisely elucidates the core risk exposure profiles across distinct operational phases throughout the entire pipeline—from token minting to terminal redemption and clearing. It systematically addresses the pivotal questions of when risks manifest, which economic agents trigger them, and where they concentrate within the operational architecture (see Table I and Fig. 2).

2) *Amplification of Stablecoin Risks: An Evolutionary Stage Perspective*: Spatially, the framework tracks a four-tier causal chain: 'Risk Origination → Vulnerability Exposure → Dynamic Amplification → Cross-Domain Contagion,' mapping atomic risks from their initial origin to cross-system spread. Through this spatial tiered mapping, the framework profoundly

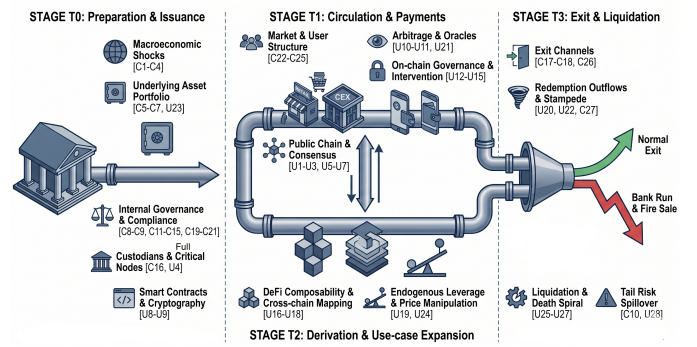


Fig. 2. Capital Flows and Risk Evolution Across the Stablecoin Lifecycle. (Note: This figure was illustrated with the assistance of Large Language Models (LLMs).)

elucidates the core logic of how idiosyncratic, localized atomic risks undergo an exponential escalation in magnitude—fueled by the endogenous structural flaws of the stablecoin system, the strategic interactions among market agents, and the breakdown of core operational mechanisms. Consequently, these risks ultimately breach the systemic boundaries of the stablecoin itself, culminating in cross-market contagion that penetrates the broader DeFi ecosystem and the legacy financial architecture. This formulation comprehensively addresses the critical inquiry of how micro-level, single-point risks progressively evolve into macro-level systemic crises (see Table II and Fig. 3).

IV. CONCLUSION AND FUTURE WORK

The core contribution of this paper lies in its systematic construction of the 55 atomic risk framework and its compre-

TABLE I
LIFECYCLE STAGES AND RISK MAPPING OF STABLECOINS

Lifecycle Stage	Phase Delineation & Scope	Key Operational Agents	Risk Mapping
T0: Provisioning & Issuance	The entire process spanning from the injection of collateral into the system to the compliant minting of stablecoins and the construction of the initial balance sheet.	Issuing institutions (e.g., Tether, Circle), decentralized smart contract protocols, and cryptographic algorithms.	• Macroeconomic Shocks: [C1-C4] • Underlying Asset Portfolio: [C5-C7, U23] • Internal Governance & Compliance: [C8-C9, C11-C15, C19-C21] • Custodian Institutions & Full Nodes: [C16, U4] • Smart Contracts & Cryptographic Algorithms: [U8-U9]
T1: Circulation & Payment	The phase wherein minted stablecoins circulate at high frequency as a pure medium of exchange within both the crypto ecosystem and real-world scenarios, absent any leverage operations.	Primary market makers, centralized/decentralized exchanges (CEXs/DEXs), retail users, payment platforms, underlying public blockchain nodes, Layer 2 scaling network sequencers, and price oracles.	• Market & User Structure: [C22-C25] • Public Blockchain & Consensus: [U1-U3, U5-U7] • Arbitrage & Oracles: [U10-U11, U21] • On-Chain Governance & Intervention: [U12-U15]
T2: Derivation & Scenario Expansion	The phase during which retails or institutional participants, having acquired stablecoins in the T1 stage, actively deploy them as financial instruments into external protocols to capture yield, thereby initiating credit derivation.	Token holders (retails), arbitrage institutions, external DeFi protocols, and flash loan attackers.	• DeFi Composability & Cross-Chain Mapping: [U16-U18] • Endogenous Leverage & Price Manipulation: [U19, U24]
T3: Exit & Liquidation	The entire process of capital exiting the stablecoin system to be redeemed for fiat currency or underlying collateral, encompassing routine redemptions, normal secondary market liquidations, as well as run-like episodes and forced liquidations under extreme stress.	Issuing institutions, redeeming users, liquidation bots, and arbitrage institutions.	• Exit Channels: [C17-C18, C26] • Redemption Outflows & Run-Driven Stampedes: [U20, U22, C27] • Liquidations & Death Spirals: [U25-U27] • Tail Risk Spillovers: [C10, U28]

TABLE II
AMPLIFICATION STAGES AND ATOMIC RISK MAPPING

Risk Tier	Tier Definition & Mechanism	Risk Mapping
L0: Initial Risk Triggers	This tier serves as the origin of stablecoin risk evolution, encompassing all exogenous shocks, native technical failures, and initial credit or compliance defaults. Operating independently of market games, these risks represent the foundational triggers of a crisis, characterized by abruptness, exogeneity, and foundational causality.	• Macroeconomic Shocks: [C1-C4] • Compliance & Credit Defaults: [C12, C15-C16, C19] • Underlying Technical Failures: [U2-U4, U8-U10] • Consensus & Cross-Chain Bridge Attacks: [U5-U7]
L1: Endogenous Systemic Vulnerabilities	This tier comprises the structural flaws inherent within the stablecoin system, serving as the core vehicles for risk amplification. While they do not independently spark a systemic crisis absent an L0 trigger, they drastically magnify the magnitude of shocks once triggered, directly dictating the evolutionary depth of the crisis. This tier is characterized by endogeneity, static nature, and structural properties.	• Asset-Side Structural Vulnerabilities: [C5-C10, U23] • Governance & Compliance Structural Vulnerabilities: [C11, C13-C14, C20-C21, U14] • Endogenous Economic Mechanism Vulnerabilities: [U24] • Fiat On/Off-Ramp & Redemption Mechanism Vulnerabilities: [C17, C26] • Market & User Structural Vulnerabilities: [C22-C25]
L2: Dynamic Market Amplification Effects	This tier represents the phase of market-driven dynamic games and operational mechanism breakdowns that occur when initial risks intersect with structural vulnerabilities. It constitutes the critical stage where risk magnitude undergoes exponential escalation. These risks manifest as rational or irrational behaviors of market agents coupled with the breakdown of systemic mechanisms under stress, transmuting localized shocks into system-wide liquidity crises and price collapses. This tier is characterized by dynamism, game-theoretic interactions, and compounding amplification.	• Operational Mechanism Breakdowns: [U1, U11-U13, U15, U18-U19, U25-U26] • Failure of Arbitrage & Stabilization Mechanisms (Pro-cyclical withdrawal of LPs): [U20-U22, C27]
L3: Cross-System Risk Contagion	This tier marks the definitive transition of a localized crisis into a global systemic crisis. The defining feature of these risks is their capacity to breach the boundaries of a single stablecoin. Through asset nesting, protocol composability, and cross-market capital velocity, risks propagate to other stablecoins, the broader DeFi ecosystem, and legacy financial markets. This tier is characterized by cross-domain propagation, systemic globality, and spillover externalities.	• Cross-Protocol Contagion Within the DeFi Ecosystem: [U16-U18] • Protocol Collapse & Cross-Market Contagion: [U27-U28]

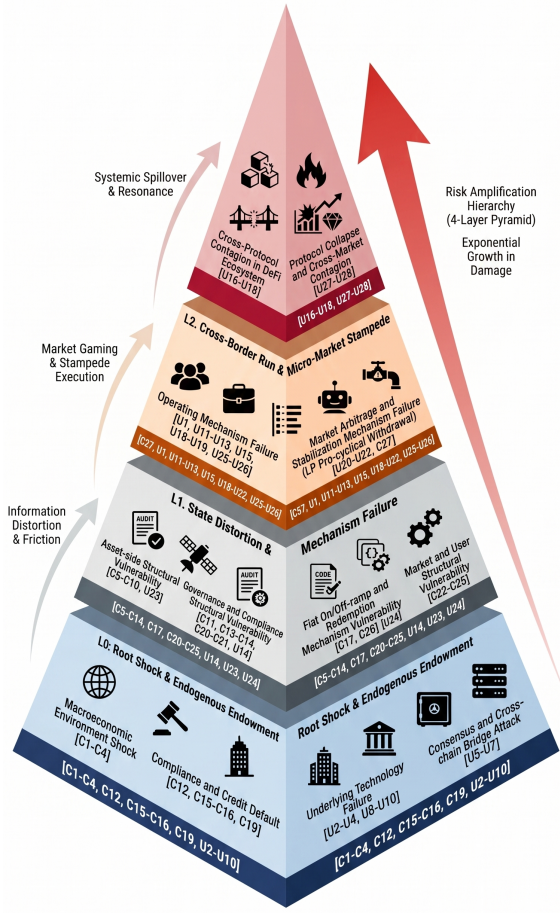


Fig. 3. The Hierarchical Pyramid of Risk Amplification. (Note: This figure was illustrated with the assistance of Large Language Models (LLMs).)

hensive risk mapping across three critical dimensions: the full lifecycle, risk amplification tiers, and regulatory frameworks. This framework effectively lays a robust theoretical foundation for subsequent comprehensive and quantitative assessments of stablecoin risks. Building upon these foundational findings and addressing extant research gaps, our future research will pivot around two core trajectories.

A. Risk Quantification and Early Warning Platform

Based on the atomic risk framework established in this paper, our future work will transition from qualitative taxonomy to quantitative modeling. We aim to develop a dynamic quantitative framework that evaluates risk transmission and compounding amplification across both temporal and spatial dimensions. Spatially, corresponding to the escalation pathway from initial triggers to cross-system contagion, we aim to quantify the amplification coefficients across distinct risk tiers. Temporally, matching the sequential operational characteristics from provisioning to liquidation, this methodology will capture the precise risk exposure profiles across various operational phases while accounting for spatial amplification coefficients. On this basis, we will construct a robust quantitative indicator matrix and a tiered early warning model to facilitate the

dynamic monitoring and prospective identification of stablecoin vulnerabilities. This system is designed to deliver predictive risk telemetry for diverse market participants—including macroprudential regulators, institutional investors, and retail token holders—thereby addressing the critical deficit in the transition from qualitative analysis to quantitative surveillance.

B. Regulatory Coverage Evaluation and Optimization Recommendations

Concurrently, future research will execute empirical evaluations regarding the risk coverage density of stablecoin regulatory frameworks across major global economies. By programmatically calculating the coverage ratios of existing regulatory regimes against the 55 atomic risks, we will analyze the precise depth of regulatory oversight across various risk dimensions within different jurisdictions. This quantitative mapping will illuminate regulatory deficiencies and governance blind spots, enabling the formulation of targeted optimization recommendations for regulatory frameworks, and ultimately providing robust theoretical benchmarks for the prudential supervision and sustainable development of the stablecoin market.

ARTIFICIAL INTELLIGENCE (AI) DISCLOSURE

In compliance with the IIKI 2026 guidelines, we disclose the use of Artificial Intelligence (AI) tools, specifically Gemini, DeepSeek, and Grok, during the preparation of this manuscript. In the preliminary stages, these tools were utilized for literature discovery and reading assistance. Furthermore, Gemini was employed to aid in the comprehension of underlying technical details (e.g., blockchain, smart contracts, and oracles), the graphical rendering of Figures 1–3 based on our explicit structural designs, as well as language translation, text refinement, and LaTeX formatting.

To rigorously ensure the accuracy, integrity, and originality of the content, we manually validated all AI-assisted outputs. All cited academic viewpoints were extracted manually through direct reading of the original source texts, rather than relying on AI summaries. Technical knowledge acquired via AI was systematically cross-referenced against authoritative literature, external reports, and textbooks. The conceptual logic and informational content of all figures were exclusively human-driven, with AI functioning solely as a rendering instrument. Finally, all refined and translated texts were meticulously audited to preserve our original intended meaning without misrepresentation. We take full and sole responsibility for the final content, originality, and findings of this manuscript.

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Due to strict page limits, the full list of references cited in this manuscript (from [1] to [25]), along with Appendix 1, are provided in our anonymized supplementary material, available for review at: https://osf.io/mx4yf/overview?view_only=9327f99d3074408482ae8beb99651ee7

Anonymized Supplementary Material for: Establishing a Multi-Dimensional Risk Framework over the Full Lifecycle of Stablecoins

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APPENDIX 1: TAXONOMY AND DEFINITIONS OF THE 55 DISCRETE ATOMIC-LEVEL RISKS

Note: As referenced in Section III.B of the main manuscript, this appendix provides the comprehensive taxonomy and detailed definitions of the 55 discrete atomic-level risks, which are categorized into Common Risks and Unique Risks.

Part A: Common Risks

Risk Dimension	ID	Atomic Risk Name	Risk Meaning
I. Macroeconomic Shocks	C1	Macro Interest Rate Shocks	Under a rate hike cycle, fixed-income reserve assets (e.g., Treasuries) lose value at mark-to-market pricing; under a rate cut cycle, interest income fails to cover operational costs.
	C2	Exchange Rate Volatility Shocks	Misalignment between the reserve asset currency (e.g., USD) and the pegged fiat currency (e.g., EUR); macro exchange rate swings cause reserve translation gaps.
	C3	Commodity Price Shocks	For stablecoins linked to commodities like gold or oil, asset value shrinks drastically due to spot market supply/demand shifts.
	C4	Extreme Geopolitical & Sovereign Shocks	Wars, sovereign debt crises, national security incidents, or sudden policy shifts cause indiscriminate dumping of safe-haven assets, freezing liquidity.
II. Structural Risks of Issuer Balance Sheets	C5	Asset-Liability Maturity Mismatch	A structural gap between the delayed liquidation of non-cash assets and the instant liquidity demand for at-par redemptions.
	C6	High Correlation of Reserve Assets	Underlying reserve assets co-crash due to extreme sensitivity to the same macroeconomic factor.
	C7	High Concentration of Reserve Assets	Capital is concentrated in too few custodian banks or specific corporate bonds, creating a deficit of risk diversification.
	C8	Deficient Profitability of Reserve Assets	Reserves mostly consist of zero-yield cash or ultra-low-yield assets, preventing the issuer from building excess reserves to absorb shocks and maintain the peg [23].
	C9	Off-Chain Reserve Re-hypothecation	To boost yields, the issuer engages in lending, pledging, or derivatives nesting of the underlying fiat deposits or sovereign bonds in traditional markets.
	C10	Insufficient Reserve Market Capacity	Underlying assets lack sufficient order book depth in traditional secondary markets, causing severe pricing discounts during mass redemptions.
III. Balance Sheet	C11	Balance Sheet Transparency Time-Lag	On-chain circulating supply changes instantly, but off-chain asset audits are delayed, concealing balance sheet deterioration.
Transparency and Internal Governance			Continued on next page

Table 1 – continued from previous page

Risk Dimension	ID	Atomic Risk Name	Risk Meaning
	C12	Snapshot Manipulation & Audit Evasion	Issuers exploit non-real-time audit loopholes by temporarily borrowing bridge funds during snapshots to hide reserve deficits.
	C13	Highly Centralized Governance Permissions	Centralized admin control over core assets or smart contracts allows illicit asset transfers, misappropriation, or financial fraud.
	C14	Lack of Bankruptcy Remoteness for Reserves	No independent statutory trust is set up; if the issuer goes bankrupt, retail user reserves are liquidated as general corporate assets.
	C15	Compliance-Driven Freeze of Fiat Accounts	Issuers fail AML/KYC checks or violate rules, prompting partner commercial banks to freeze fiat accounts.
IV. Custodians and Settlement Infrastructure	C16	Custodian Credit Default	Custodian banks face insolvency, commercial paper issuers default, or derivative counterparties fail to deliver, causing losses to reserves.
	C17	Fiat Clearing Temporal Mismatch	Crypto operates 24/7, but traditional banks close during weekends/holidays; this trading time gap amplifies de-pegging under stress.
	C18	Fiat Clearing Network Channel Congestion	Major clearing networks (e.g., Fedwire, SEN) clog or go down, blocking minting/redemption and shrinking secondary market liquidity.
V. Regulatory Compliance and Licensing	C19	Regulatory Non-Compliance Default	Daily operations fail local statutory requirements, leading directly to fines, mandatory liquidation, or license revocation.
	C20	Low-Tier Licensing Credentials	The issuer lacks high-standard regulatory backing or operates in a weak jurisdiction, leaving token holders with insufficient legal protection.
	C21	Regulatory Heterogeneity	Fragmented rules across jurisdictions cause regulatory arbitrage, spatial liquidity division, and high compliance costs, sparking disorderly runs and lack of cross-border resolution.
VI. Micro Market Structure and Token Holder Behavior	C22	Deficient Secondary Market Order Book Depth	Order book depth is thin and often masked by wash trading; risk models overstate system resilience, so small sales trigger huge price slippage.
	C23	Highly Concentrated On-Chain Token Holdings	Circulating supply is concentrated in a few “whale” addresses; under stress, their liquidations easily overwhelm market capacity.
	C24	User Structural Vulnerability	Tokens are held primarily for speculation rather than day-to-day payments, meaning low-stickiness users easily panic and trigger stampede runs.
	C25	Erosion of the Credit Foundation	The price peg relies heavily on user confidence; past de-pegging, negative audits, or macro stress damage trust, making the stablecoin permanently run-prone.
	C26	Asymmetric Redemption Channels	Institutions can redeem 1:1 frictionlessly in primary markets, while ordinary retail users are restricted to secondary AMM pools, inciting retail panic.
	C27	Homogeneous Algorithmic AI Agent Runs	Robo-advisory AI agents using similar data and models execute the same liquidation commands simultaneously upon capturing minor de-pegging signals.

Part B: Unique Risks

Risk Dimension	ID	Atomic Risk Name	Risk Meaning
I. Blockchain Infrastructure and Network Consensus	U1	Underlying Public Blockchain Network Congestion	The public blockchain hits its bandwidth limit or crashes, blocking transactions, trapping ordinary users, and preventing liquidators from closing positions.
	U2	Layer 2 Sequencer Downtime	The centralized sequencer of a scaling network fails, halting L2 liveness and freezing tokens for hours or days.
	U3	Layer 2 Prover Logical Vulnerability	Mathematical logic flaws in the ZK Prover submit invalid transaction states to Layer 1 or permanently trap funds in cross-chain contracts.
	U4	Critical Full Node Downtime	Traffic heavily concentrates on block-producing nodes or RPC endpoints, causing single points of failure (SPOF) that halt transaction processing.
	U5	P2P Network Node Isolation	Attackers use Sybil or Eclipse attacks to isolate honest nodes at the network communication layer, blocking genuine transaction broadcasting.
	U6	51% Attack and Block Reorganization	Malicious validators hijack over half the network power to roll back transaction history, breaking instant finality and causing double-spending.
	U7	Cross-Chain Bridge Mapping De-pegging	The bridge smart contract pool is hacked or glitched, causing mapped tokens on the destination chain to lose pristine asset backing.
II. Protocol Code, Cryptography, and Governance Permissions	U8	Smart Contract Vulnerabilities	Reentrancy, integer overflow, or state variable design flaws allow hackers to drain collateral or mint unauthorized tokens.
	U9	Cryptographic Algorithm Compromise	Existing asymmetric encryption (e.g., RSA, ECDSA) is mathematically solved or broken by sudden technological leaps like quantum computing, destroying on-chain asset ownership verification.
	U10	Oracle Data Source Distortion	External spot market APIs or exchange feeds encounter outages or provide erroneous price data, writing distorted valuations onto the ledger.
	U11	Oracle Latency	Extreme network congestion delays oracle price updates during market crashes, allowing users to exploit stale, artificially high prices.
	U12	Absence of Emergency Pause Mechanisms	Certain decentralized stablecoins lack pause mechanisms to manually halt operations during code exploits or oracle failures. [Applicability: Purely crypto-collateralized and purely algorithmic models]
	U13	Contract Pause Inducing Arbitrage Disruption	Centralized contracts trigger a global pause to block hackers, which halts primary minting/redemption channels and stops market makers from closing price gaps. [Applicability: Fiat-collateralized and hybrid models]
	U14	On-Chain Blacklist Address Freezes	Issuers invoke contract blacklist functions to lock specific addresses for compliance, creating a credit split between “clean” and “tainted” assets. [Applicability: Fiat-collateralized and hybrid models]
III. DeFi Nesting Composability and On-Chain Micro-Games	U15	DAO Governance Proposal Manipulation	Attackers buy up governance tokens or use flash loans to instantly capture voting power and forcefully pass malicious proposals.
	U16	Cross-Token Cross-Collateralization Contagion	Hybrid stablecoins use dominant fiat stablecoins as reserves; if the underlying stablecoin de-pegs, it directly shrinks the asset backing of the upper stablecoin.
	U17	AMM Liquidity Pool Concurrent Contagion	Multiple stablecoins share an AMM pool; when one token faces extreme sell-offs, the constant product math drains the liquidity of healthy tokens.
	U18	DeFi Nested Collateralization Risks	Stablecoins serve as collateral in external protocols; their price drop triggers cascading liquidations, while external protocol failures erode stablecoin reserves.
	U19	Flash Loan Oracle Manipulation	Attackers use massive uncollateralized flash loans to crash spot pools on DEXs, distorting TWAP prices to trigger faulty global liquidations.
	U20	Pro-Cyclical Withdrawal of LPs	On-chain DEXs rely heavily on liquidity providers (LPs) but lack statutory market makers; during panics, LPs withdraw capital pro-cyclically.
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Table 2 – continued from previous page

Risk Dimension	ID	Atomic Risk Name	Risk Meaning
	U21	High Friction Inducing Arbitrage Failure	Network congestion drives up Gas fees; when arbitrage fees exceed de-pegging profits, market makers halt operations, disabling price stabilization.
	U22	MEV Transaction Ordering Front-Running	On-chain validators and bots front-run user orders via transaction sorting rules, extracting slippage value and amplifying price volatility.
IV. Protocol Endogenous Economic Mechanisms and Tail Risk Contagion	U23	Intrinsic High Volatility of Collateral Assets	Underlying crypto collateral assets lack macro-economic value anchoring and undergo extreme plunges during liquidity contractions.
	U24	Excessive Credit Expansion via AMO Algorithms	The protocol uses AMO modules to mint tokens directly into DEXs, creating massive shadow leverage that triggers a death spiral during a run and governance token collapse.
	U25	Unbuffered Cascading Liquidations	On-chain lending protocols liquidate via automated code without circuit breakers; dropping below safety thresholds triggers a cascading negative feedback loop.
	U26	Collateral Auction Failures from Rigid Thresholds	Mathematical rigidity in contract liquidation parameters (e.g., fixed discounts) makes liquidations unprofitable during fast price crashes, causing auctions to fail.
	U27	Dual-Token Death Spiral	To maintain the peg during a run, the system hyper-inflates its governance token to absorb sell pressure, crashing the token price and destroying the entire protocol.
	U28	Cross-Market Risk Spillovers	A plunge in on-chain crypto collateral triggers liquidations and redemption runs, forcing issuers to fire-sell assets in traditional financial markets, spilling risk into the macroeconomy.